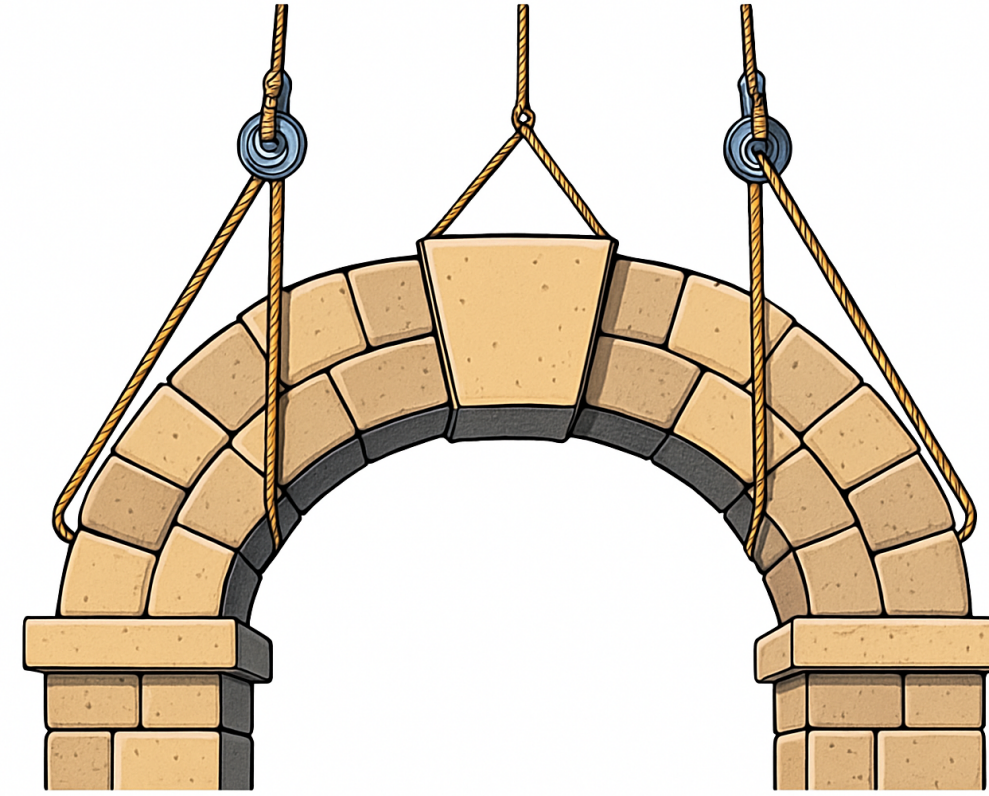




The Keystone Project

Data driven uncertainties for neutrino cross sections

Alexandra Trettin, Colton Hill
NuWro Workshop
16.09.2025

*very very
much WIP!*

Motivation

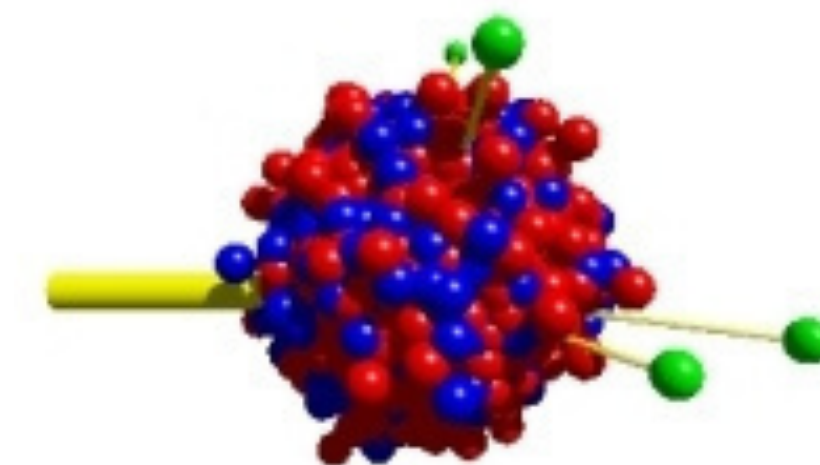
Global analyses need global systematics

Challenges in modelling xsecs

- No exact solution to nuclear interactions, only different ways of approximation
- Different MC generators making different assumptions

Cross section uncertainties?

- Uncertainties only exist for GENIE
 - GENIE “tune”: Version of GENIE with parameters adjusted to a particular measurement
- ➔ Lot of effort to produce one tune, tons of MC required

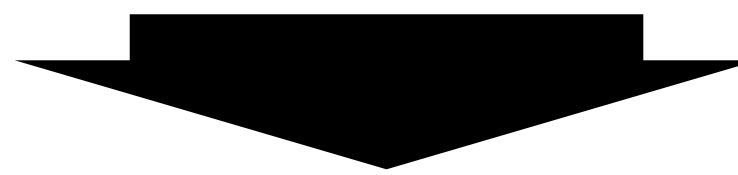
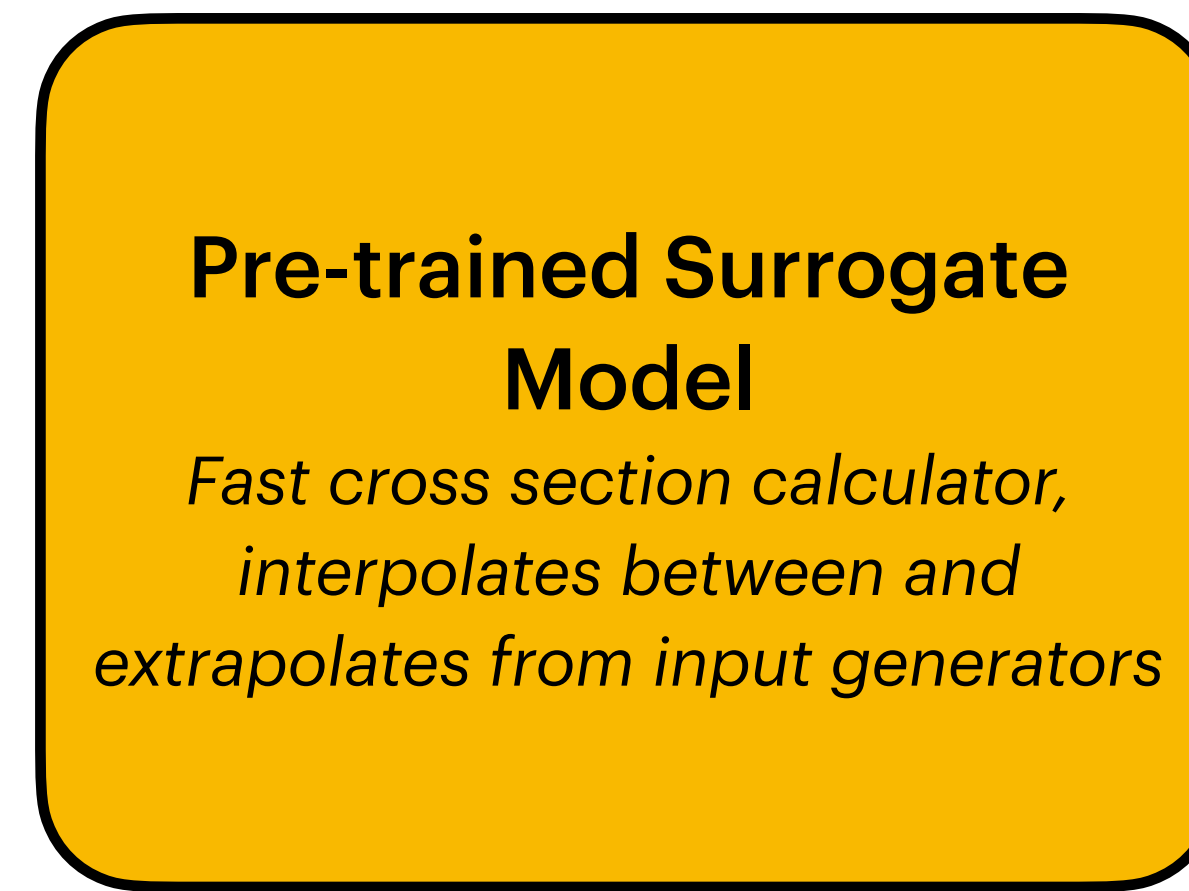
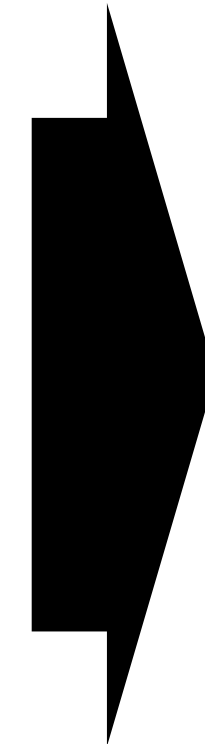
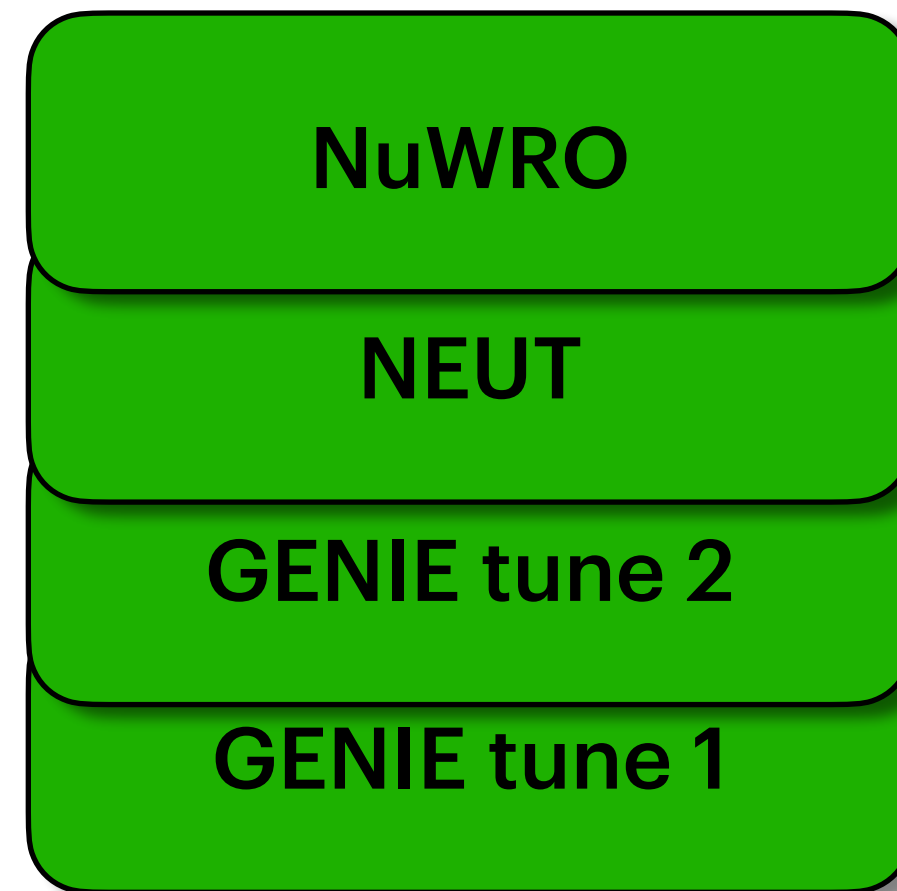


Missing link: A coherent way of quantifying and applying uncertainties across all generators, all measurements, without needing massive amounts of MC!

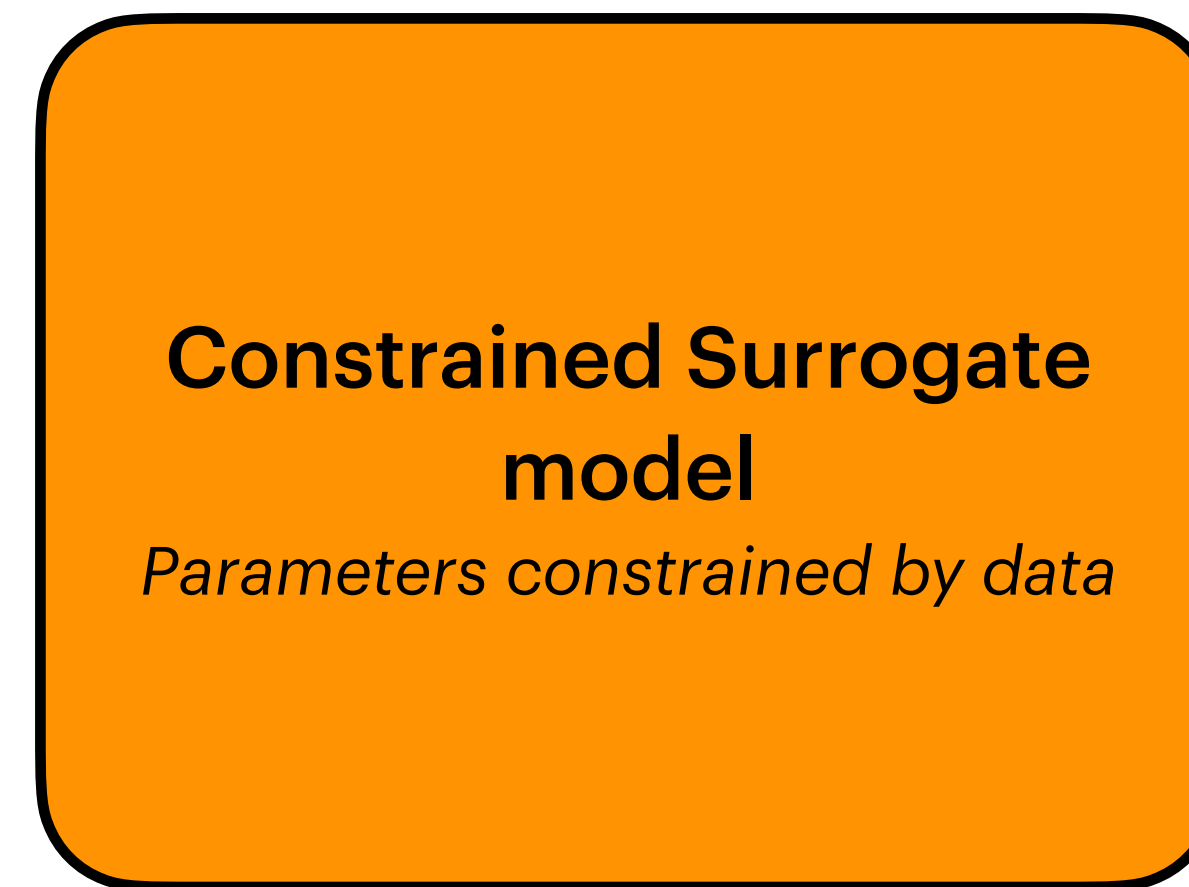
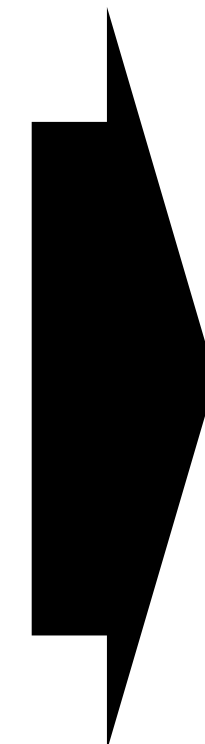
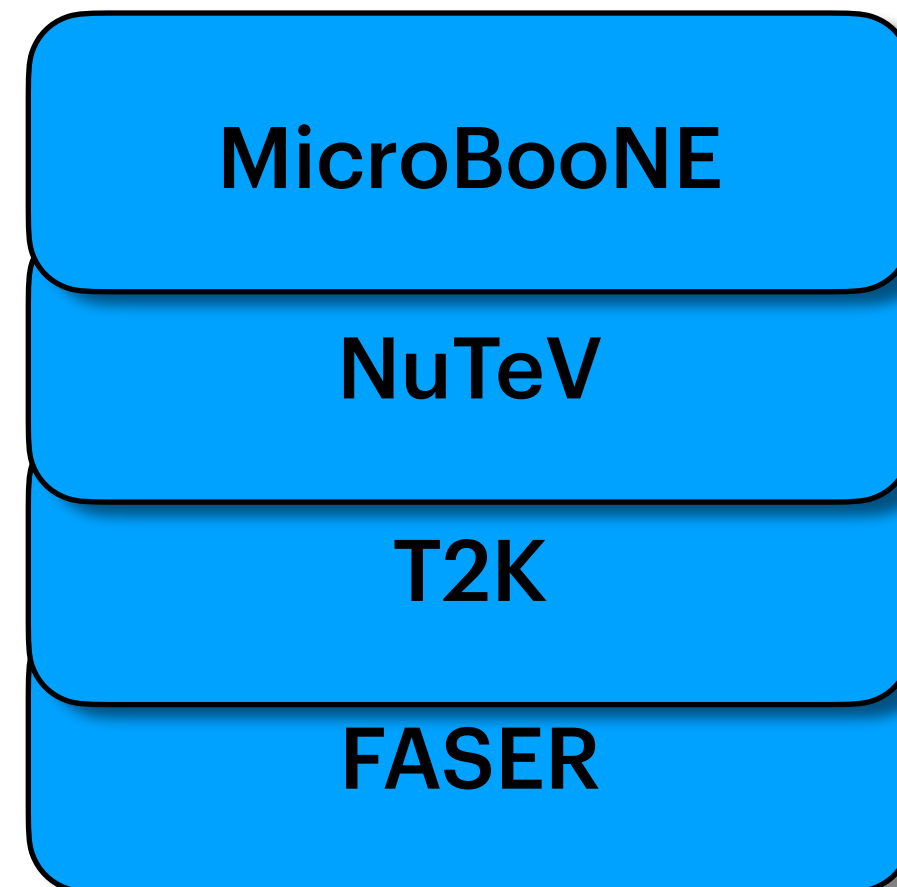
Overview

The full workflow

Generators



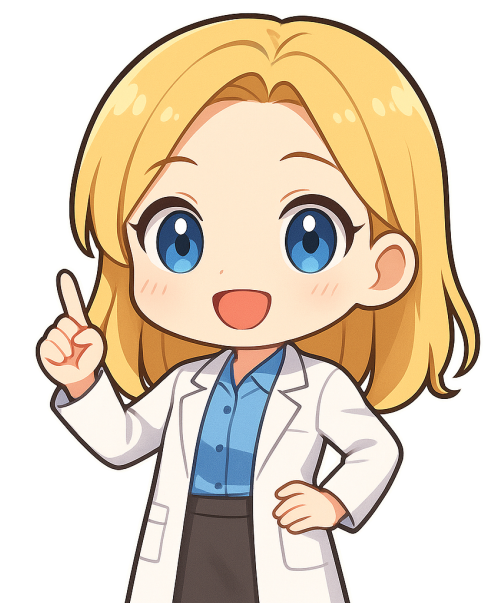
Measurements



Constrained model capabilities

- Calculate cross sections (fast!)
- Calculate robust uncertainties
- Sample “universes”
- Reweight ν events based on kinematic parameters

The constrained surrogate model is our main deliverable!



Collecting Predictions

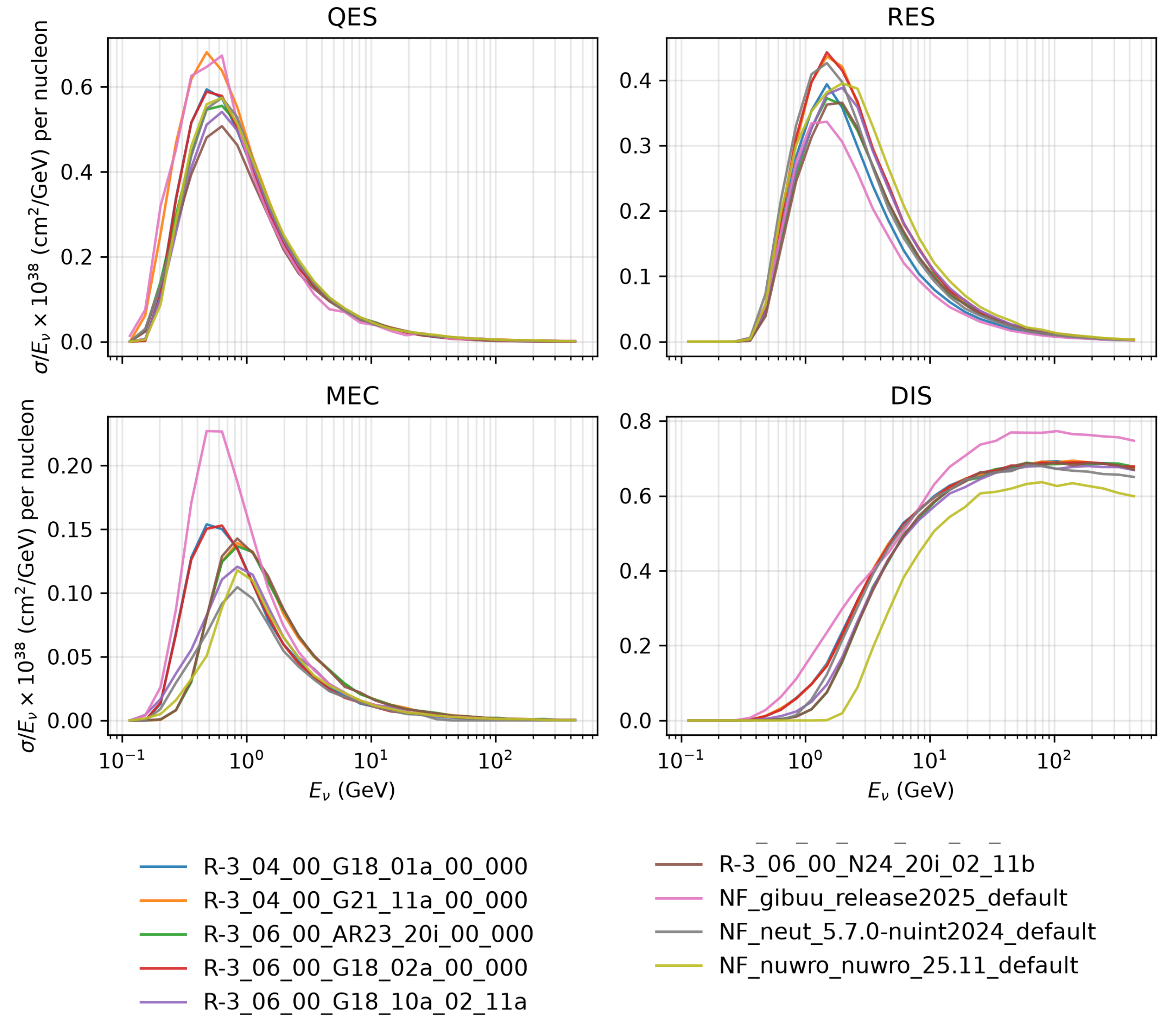
Included Generators

- 6 GENIE tunes
- GiBUU (release 2025)
- NEUT 5.7
- NuWro 25.11

Cross section extraction

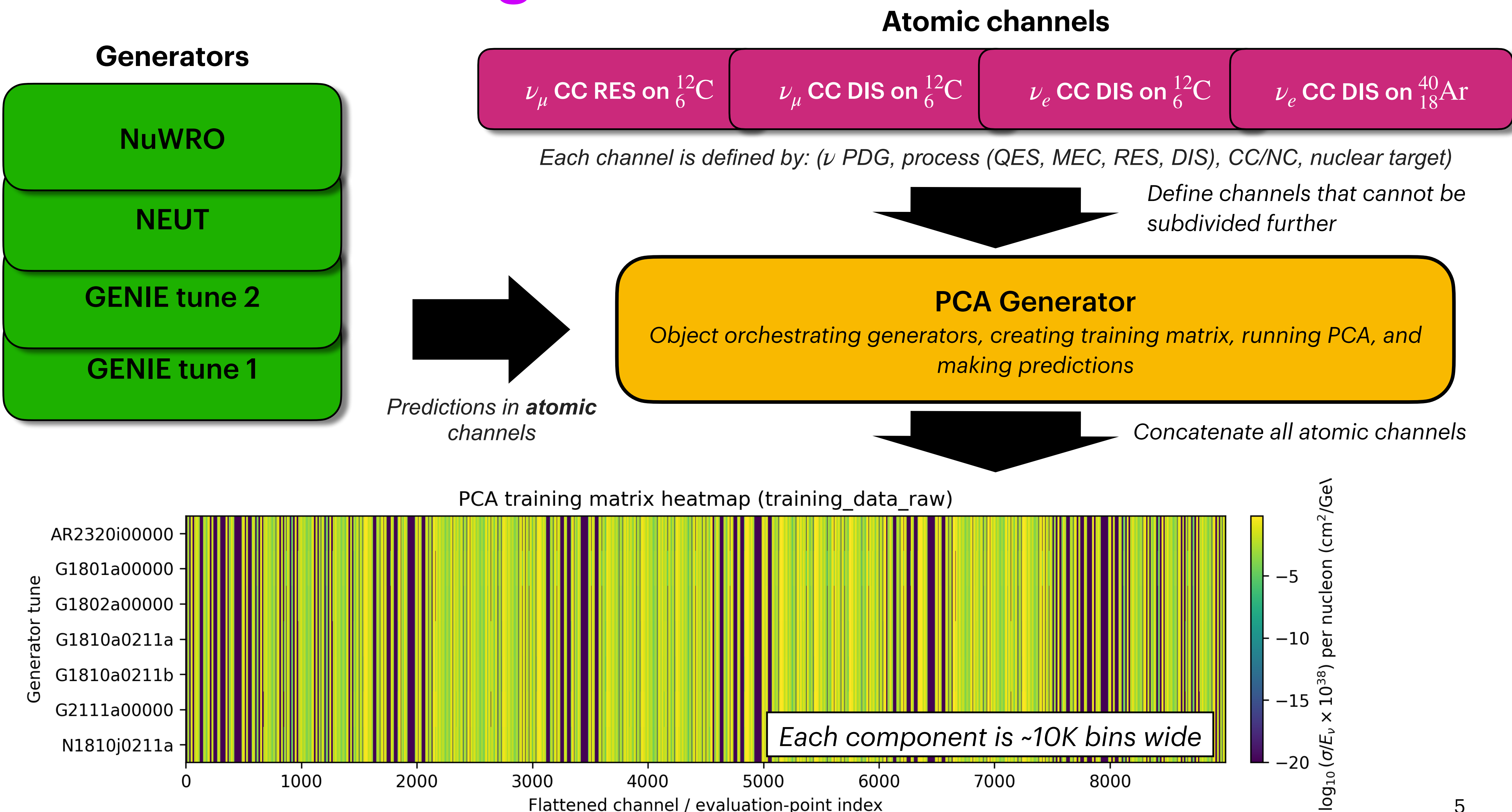
- Synthetic flux: $\Phi(E) \propto E^{-2}$ in energy range from 0.1 GeV to 500 GeV

- Density histogram with event weights equal to $w_i = \frac{1}{\Phi(E_i)}$



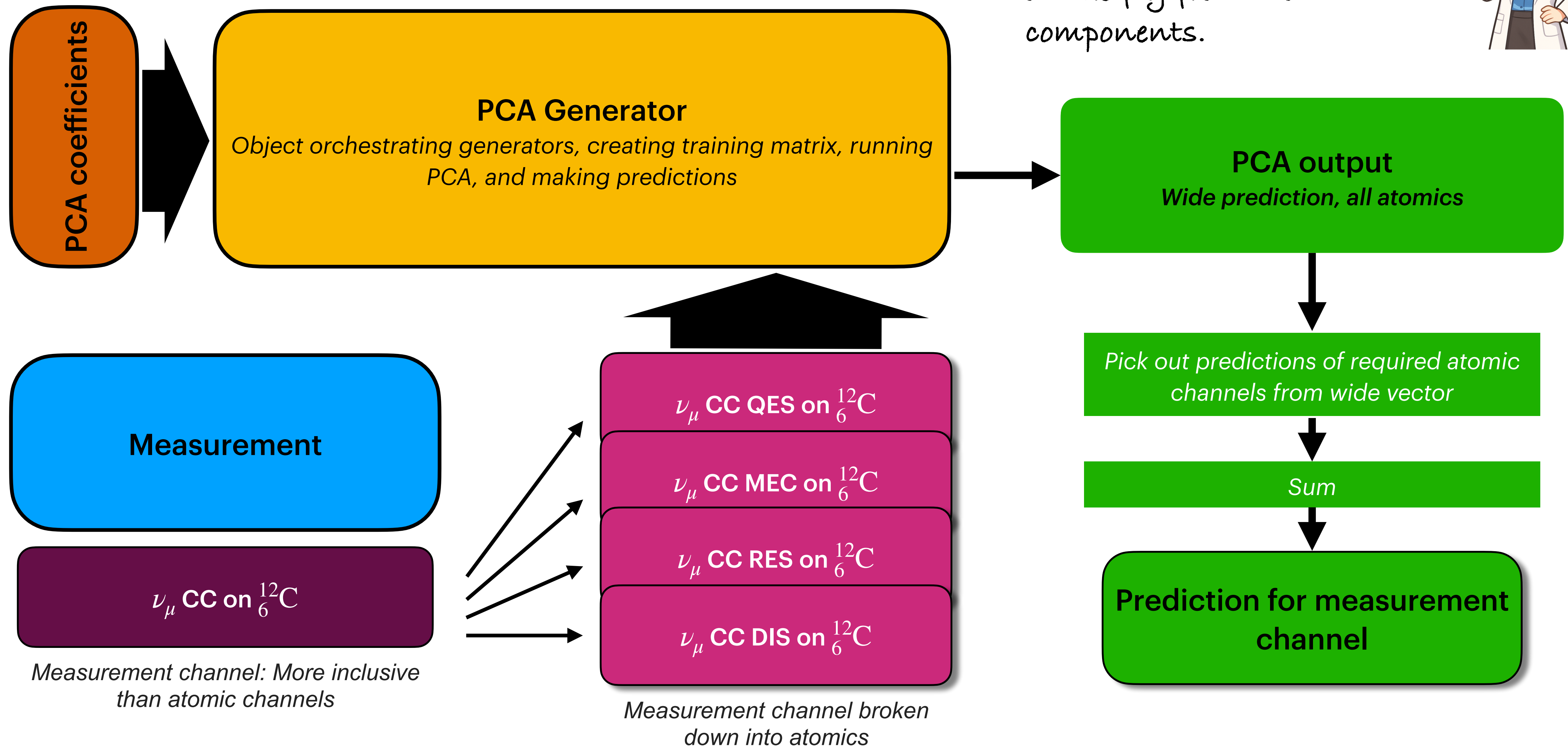
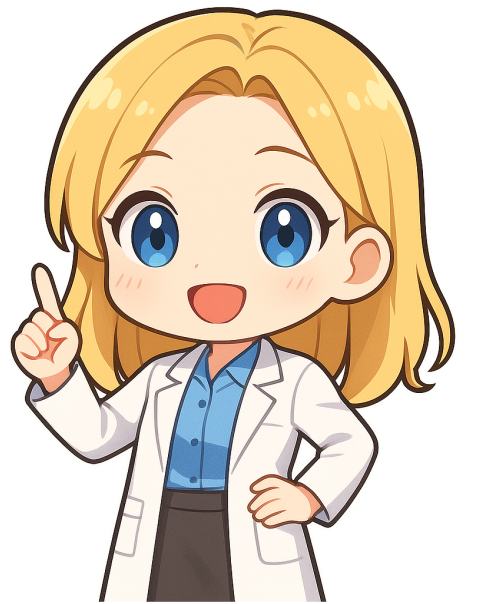
Predictions of the total cross section for each generator and GENIE tune included in the fit

A PCA-Based Surrogate model

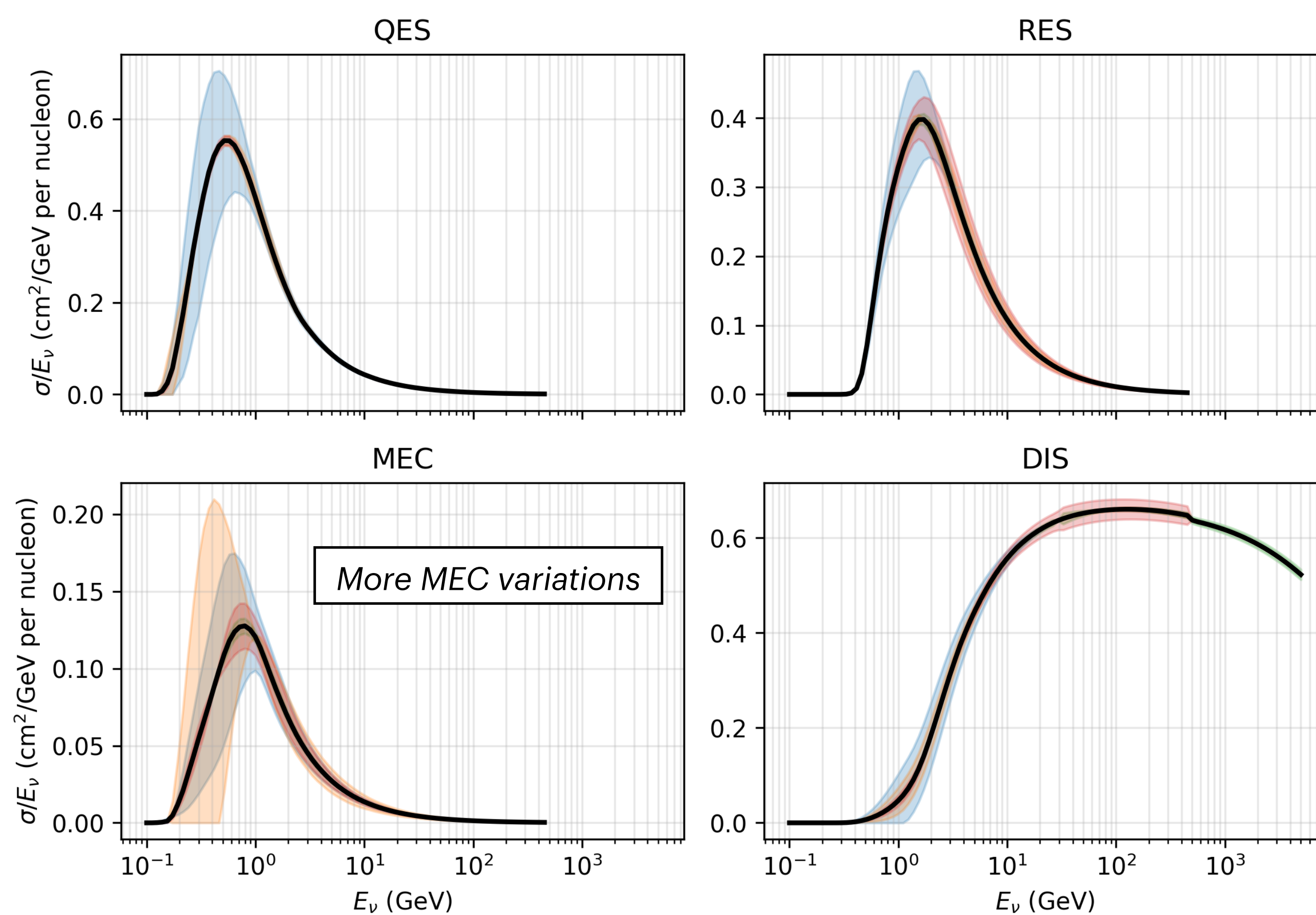


PCA Evaluation

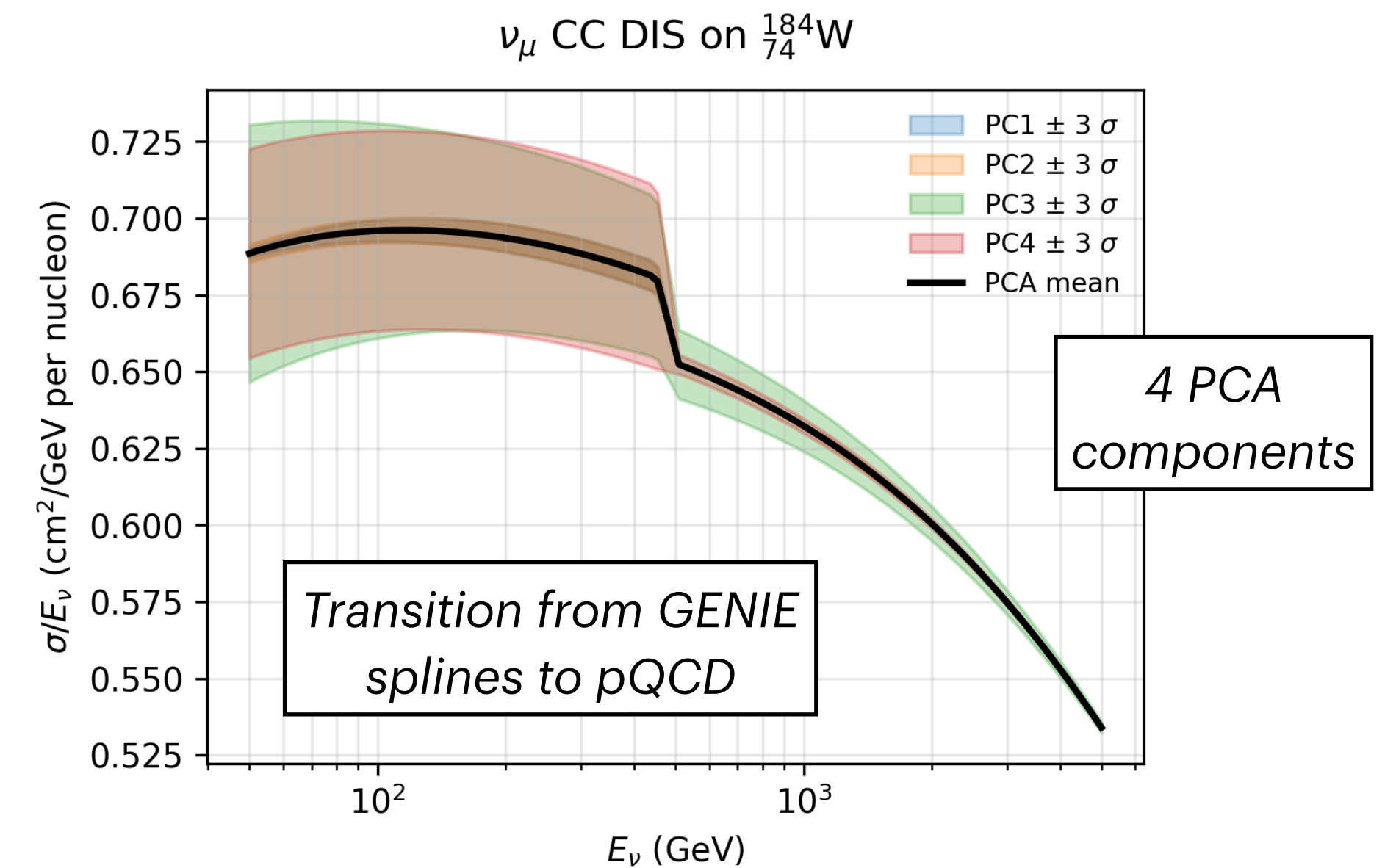
More complicated derived channels, such as averages and ratios, can be generated on the fly from atomic components.



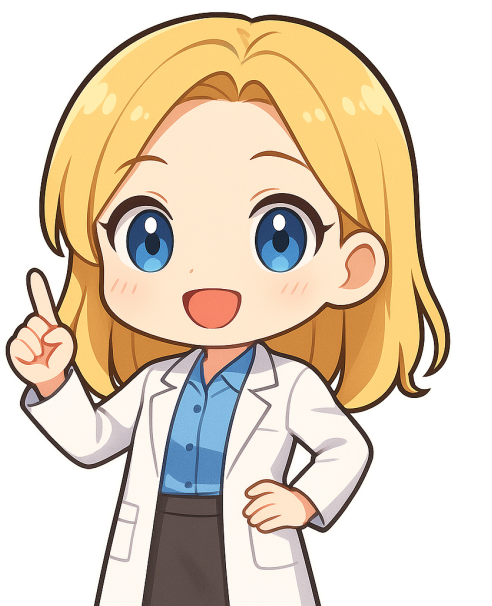
Extending the Energy Range: GENIE + pQCD



Predictions: ν_μ on carbon

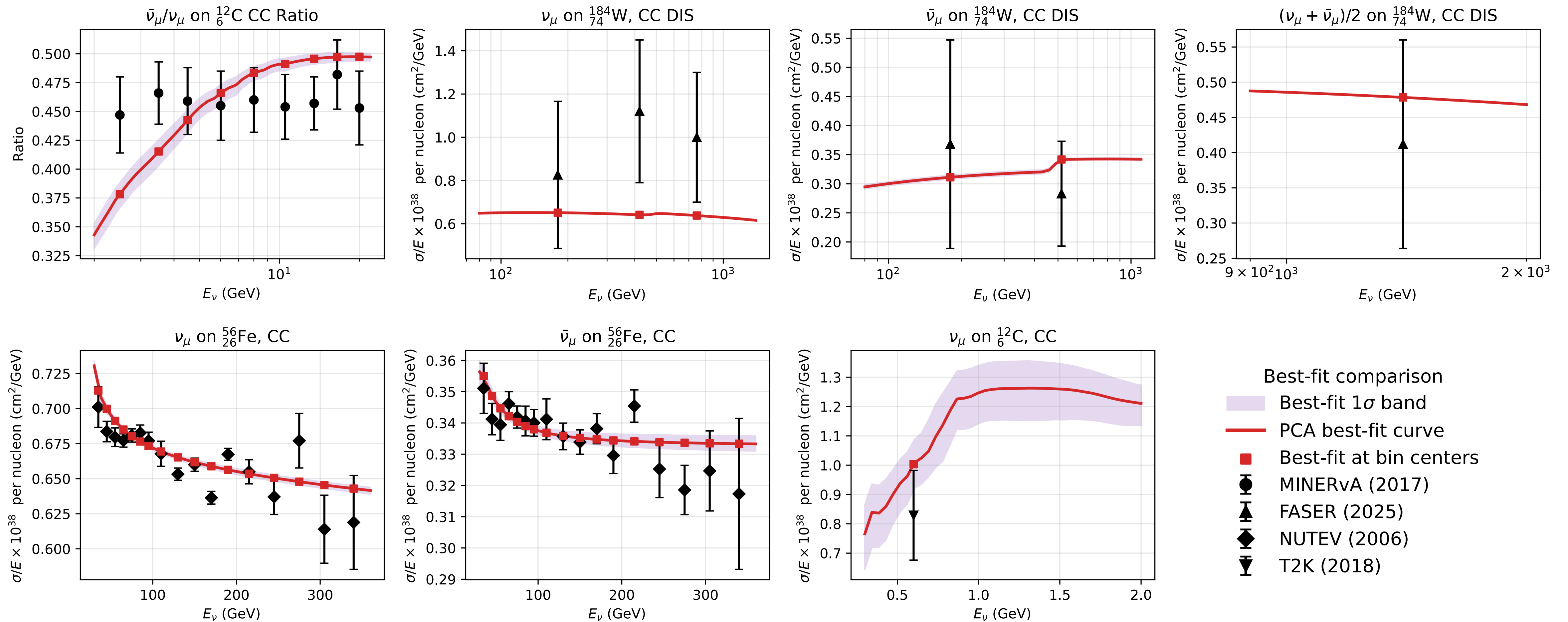


This earlier version of our model used only GENIE splines from different tunes, and combined them with perturbative QCD DIS models (WCG, CSMS) above 500 GeV.



Fitting Multiple Experiments

First results across a large energy range



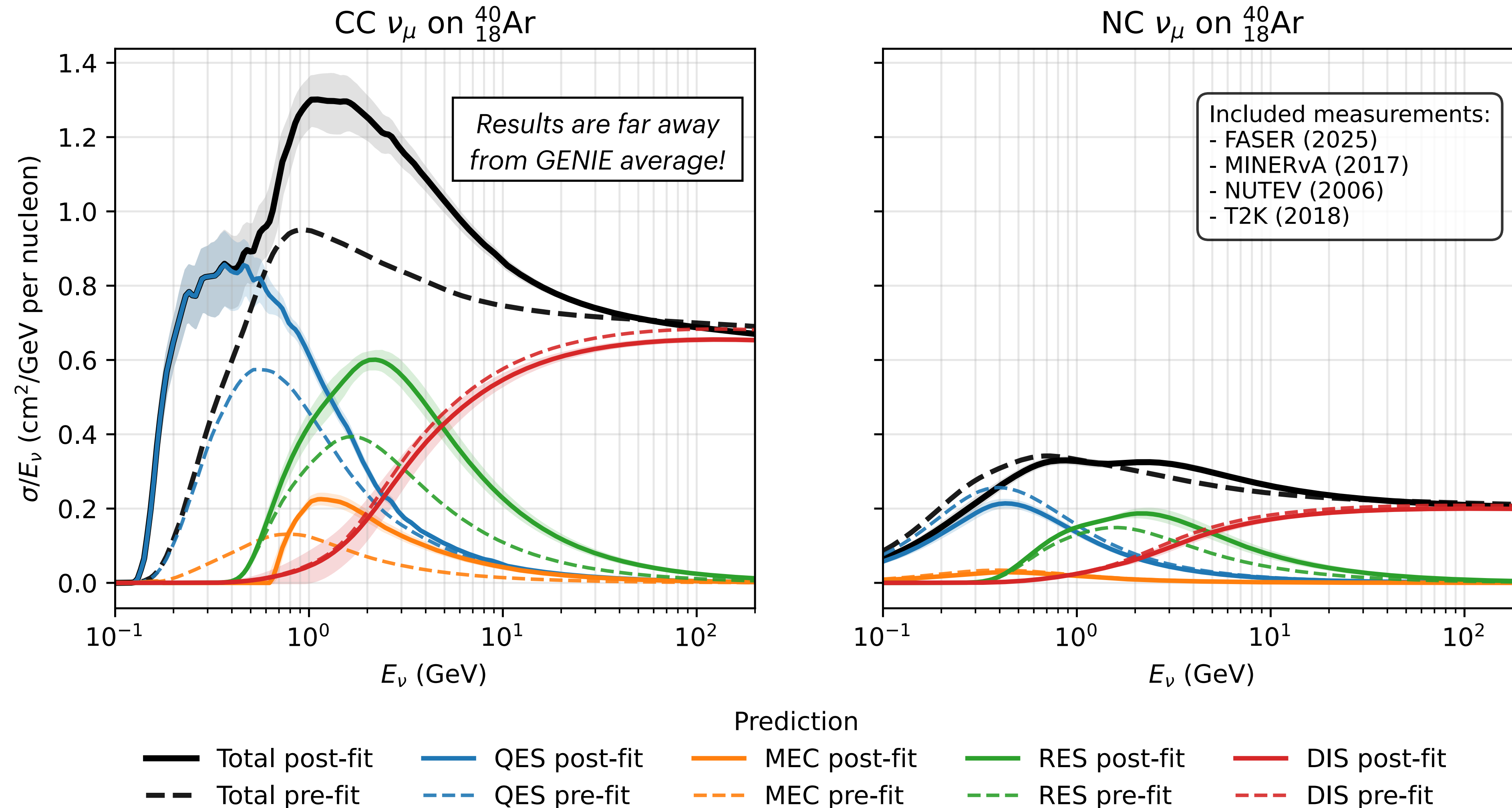
Overall: $\chi^2/\text{dof} = 182/46 = 4.9$

Fitting Multiple Experiments

Predicted cross sections after fit



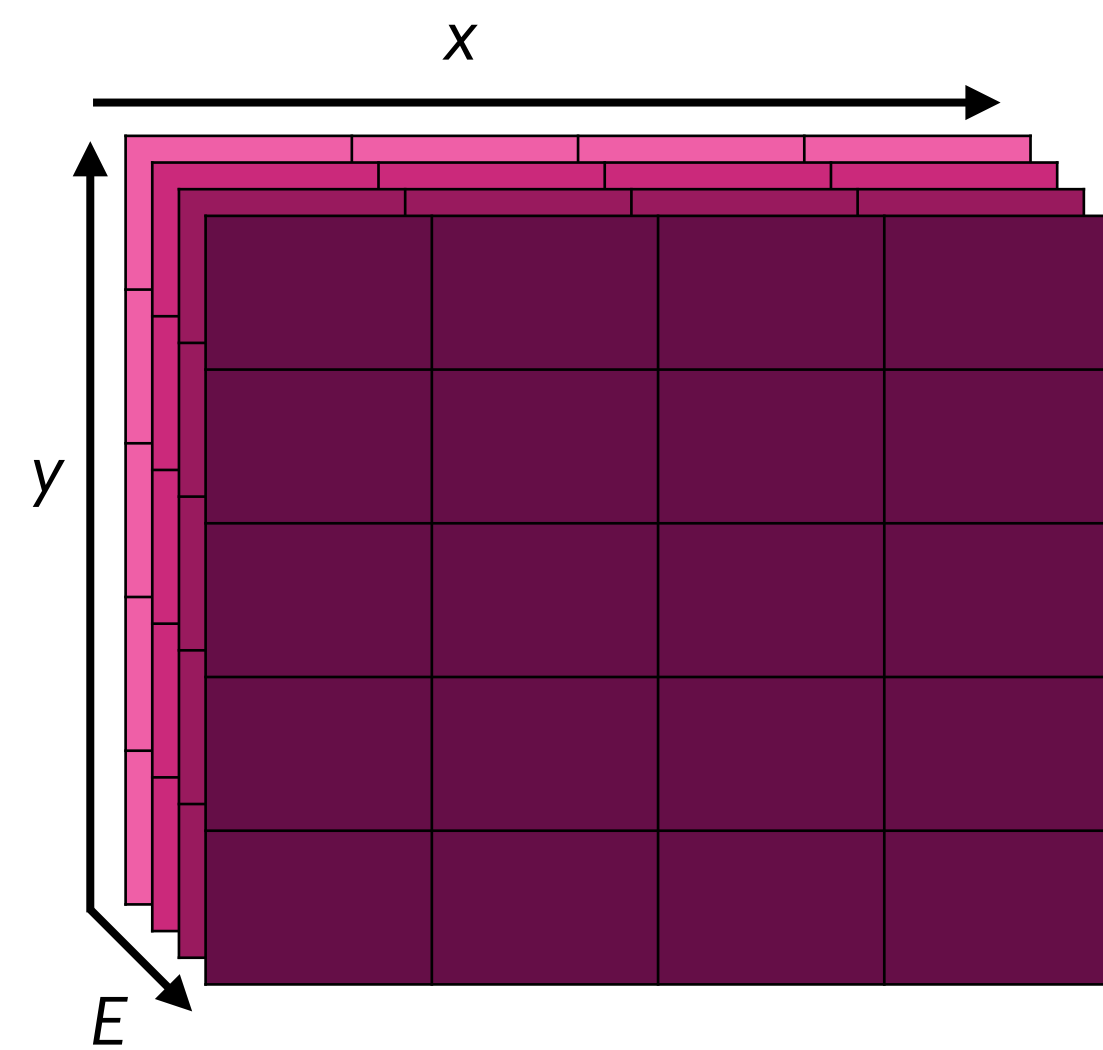
Extremely
early work!
Take these
cross sections
with a big
grain of salt!



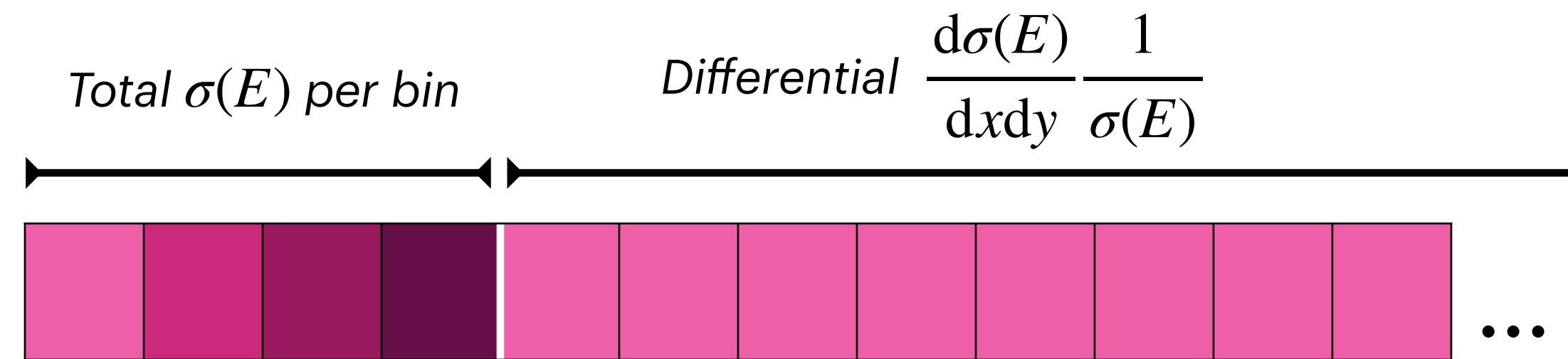
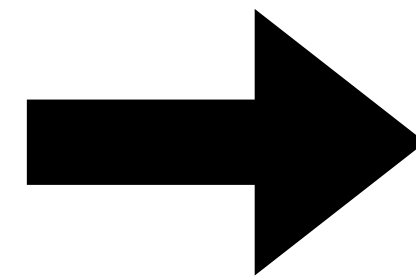
Differential Cross Sections

Our reparametrization:

- Starting point: ND density histogram of MC events → Make choice of variables depending on experimental result (for example, Bjorken-x, inelasticity-y)
- Reparametrize into **total** and **shape**, grouped by energy
- Unroll into 1D: One block for totals, one block for shapes → feed into PCA as before



Density histogram in E , x and y



- Normalization: All differential bins add to unity
- Completeness: Full kinematic range covered, sum over bins = cross section integral

Fitting Differential XSecs

Data: NuTeV – PRD 74, 012008 (2006)

- Provides measurements binned in (E, x, y) with full covariance matrix
- Covers critical energy region between low-energy generators and DIS models

Fit model: PCA over MC generators

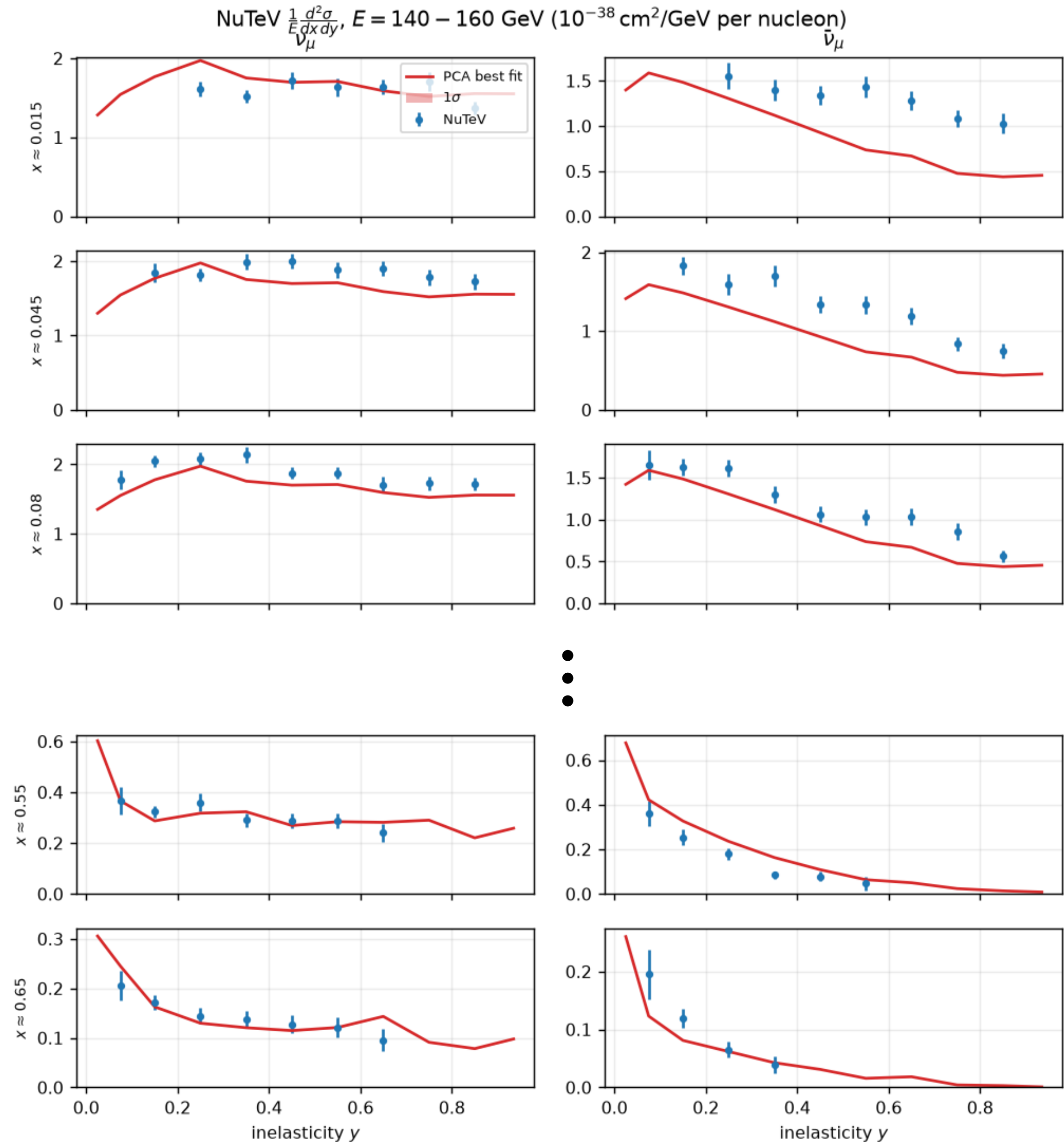
- Using MC from 6 GENIE tunes, NuWro, GiBUU, and NEUT
- Surrogate model: PCA with differential cross sections in (E, x, y), 8 components

Fit results

- Shape agrees reasonably well with data!

$$\chi^2/\text{dof} \approx 7.6$$

- Prediction errors tiny: Model overconstrained by many data points



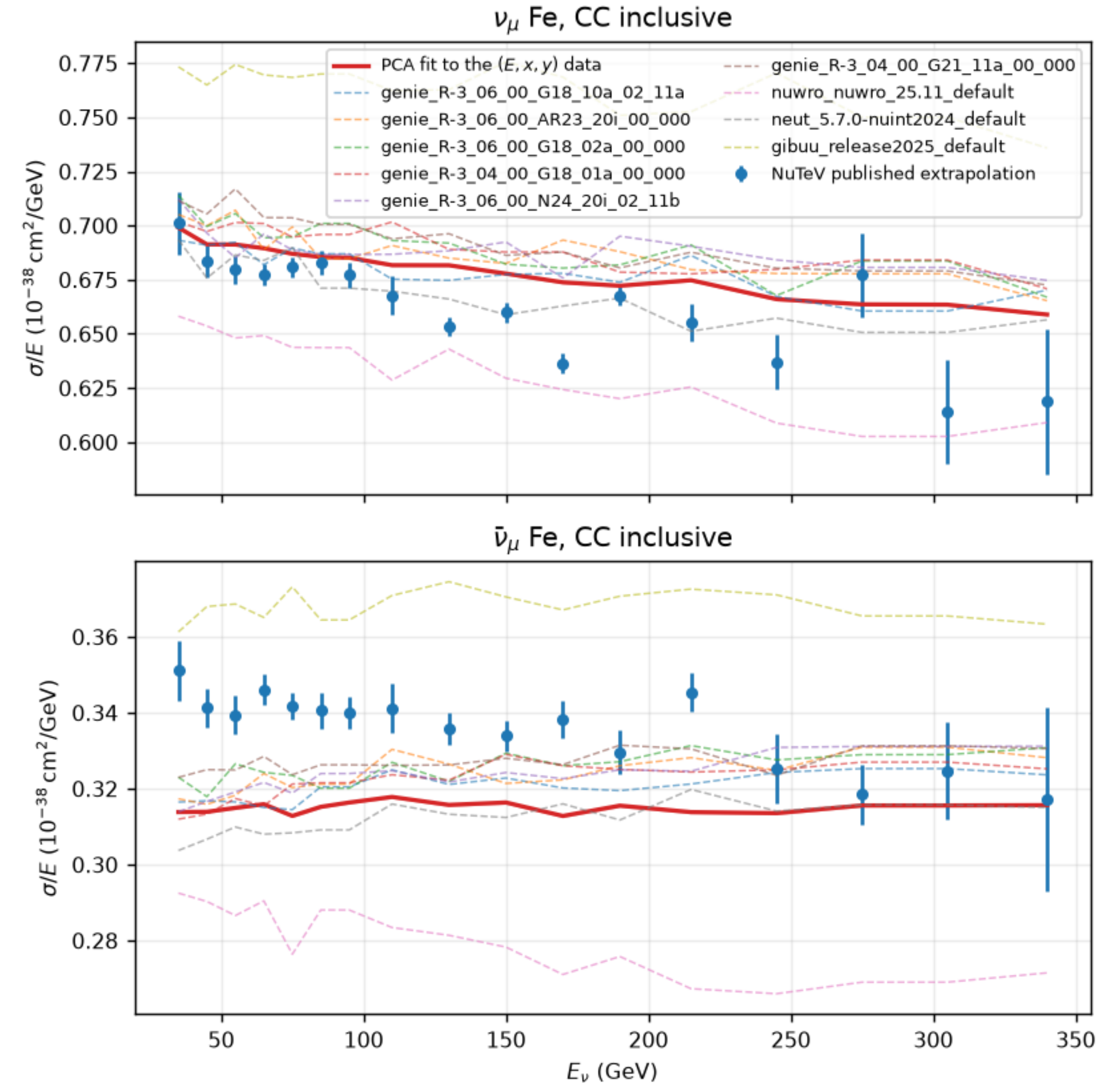
Fitting Differential XSecs

Extrapolation to total cross section (Paper)

- Detector does not observe full phase space
 - ➔ Extrapolation necessary to perform integral, paper makes simplifying assumptions
 - ➔ Least phase space measured at low E_ν , most extrapolation, least trustworthy

Our method

- Surrogate model natively gives total cross sections
 - ➔ Can simply use the constrained model to predict total
- Our results disagree with paper at low energies



Total cross sections published by NuTeV collaboration (error bars) compared to the event generator predictions (dashed lines) and the PCA best fit (red).

Planned Deliverables

What will YOU get from us?

Constrained Surrogate Model

- Easy to use, fast (millions of events per second) cross section calculator
- Constrained with our selection of data
- Reweighting tool for existing MC sets supporting “atmospheric style” (few-parameter iterative regression) and “accelerator style” (throw universe weights) analyses

Experiment Data Browser

- Look at public data we have prepared for fitting, get metrics for how compatible different data sets are with one another
 - ➡ We will include as many publications as possible!

Build-your-own-tune Tool

- Choose what measurements you want to tune to, create constrained model on the fly

Status and Outlook

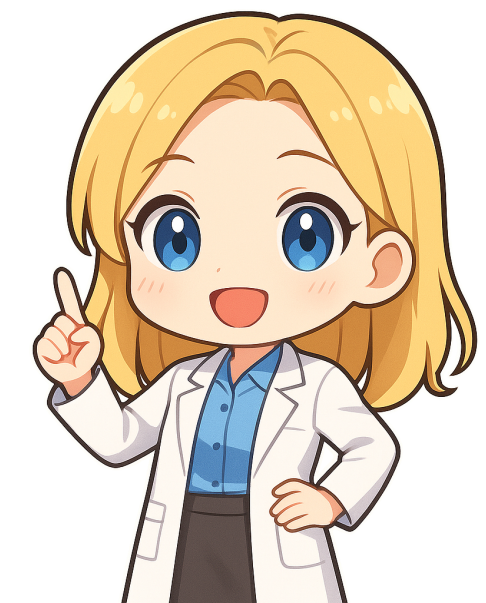
What works now

- ✓ Can fit PCA surrogate model covering $\sigma(E)$ for wide energy range
- ✓ Fitting differential cross section measurements
- ✓ Fit multiple experiments at the same time
- ✓ Most surrogate model capabilities (predict unseen channels, throw samples)

Future plans

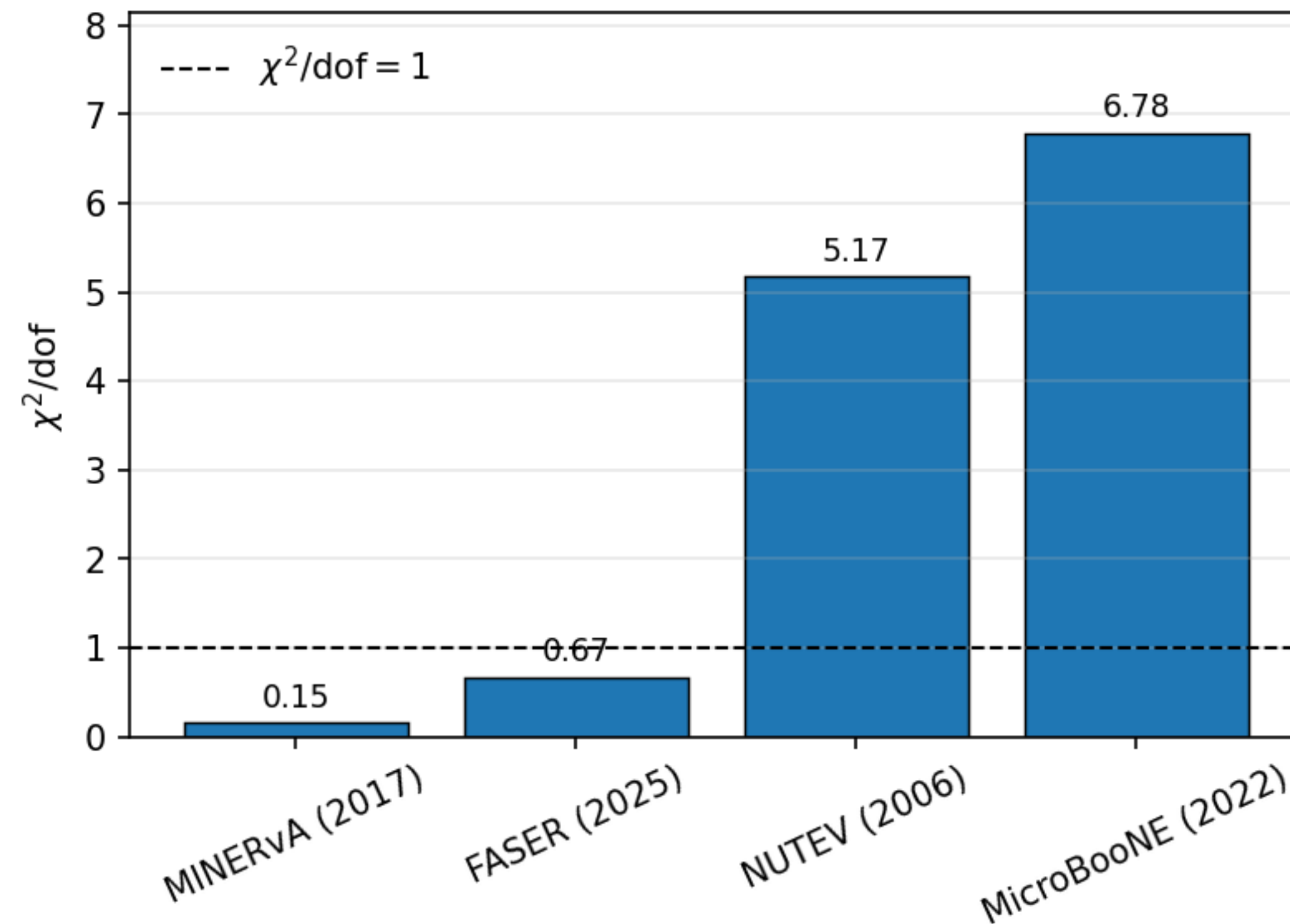
- Differential cross sections at higher DIS energies
- Support measurements in reconstructed space
- First version of a constrained surrogate model later this year!

This is very early work! We have a proof-of-principle, but the mechanics of the framework can and will change.

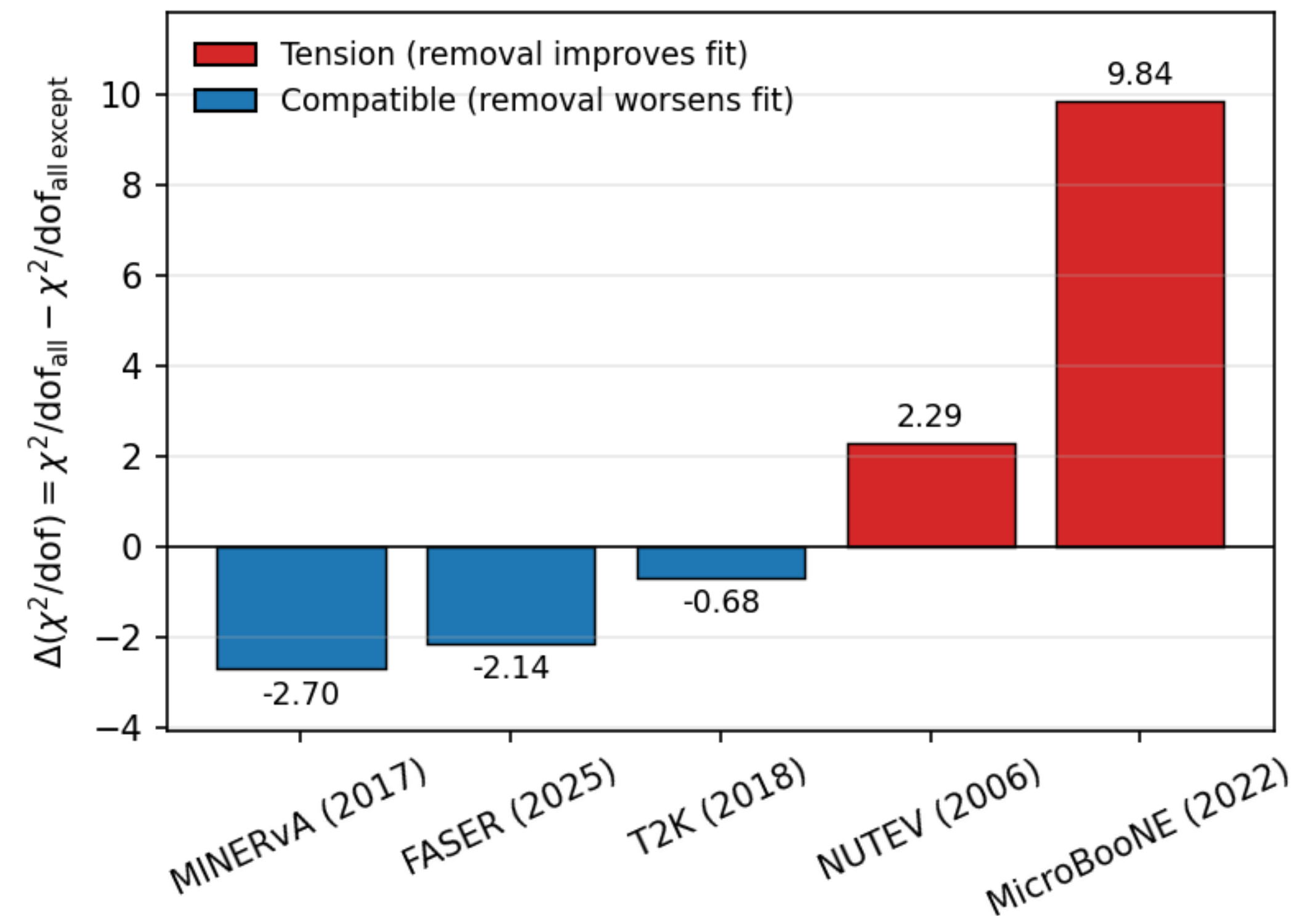


Backup

Fit Quality Diagnostics



Fit quality for individual experiments.

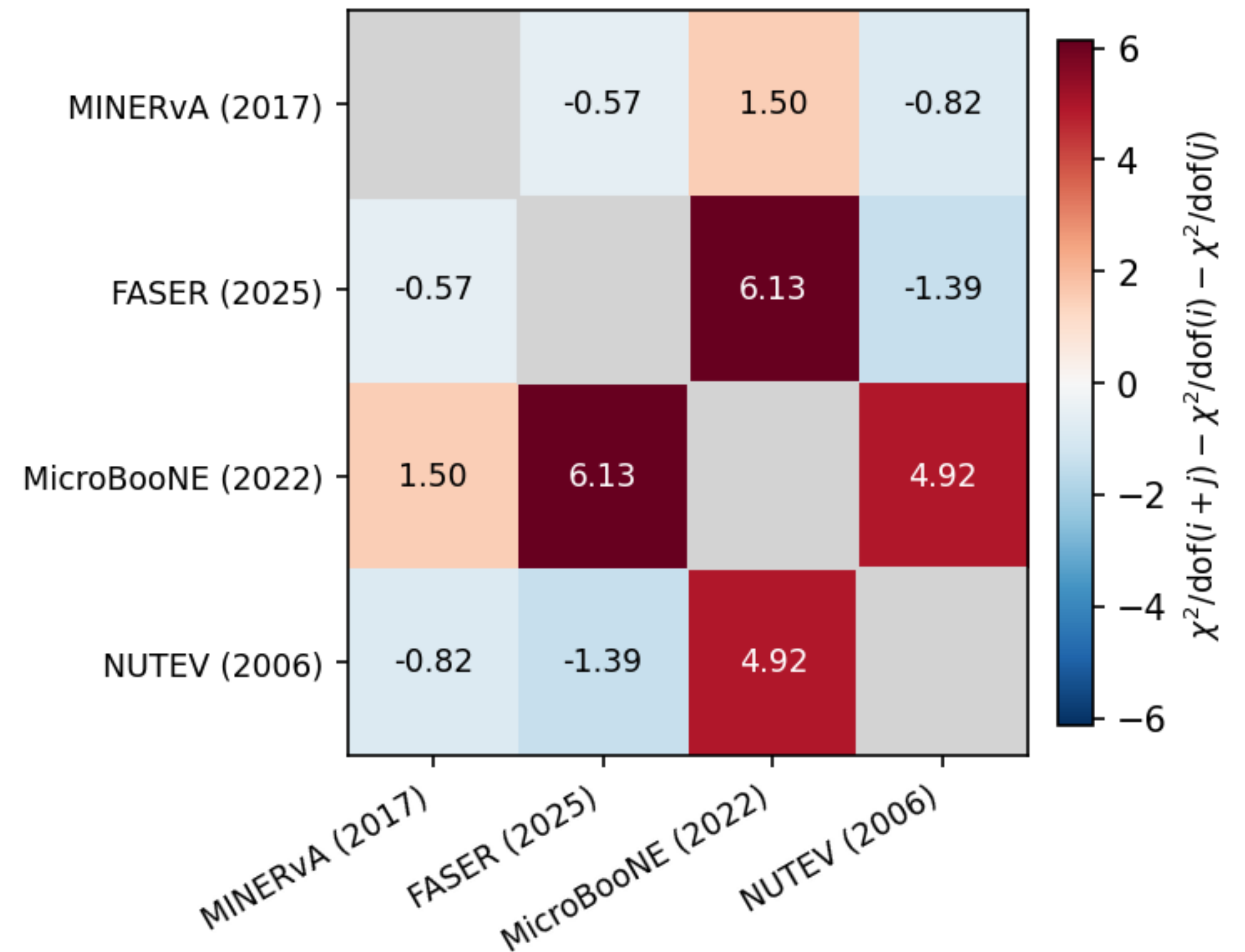


Leave-One-Out compatibility metric: Does taking an experiment out of the joint fit improve or worsen the fit quality?

Fit Quality Diagnostics

Pairwise Fit Quality Assessment

- For each pair of experiments, difference between χ^2/dof of joint fit and sum of individual test statistic values
 - ➡ Negative value: Joint fit better
 - ➡ Positive value: Joint fit worse
- MicroBooNE fits badly with everything else
 - ➡ DIS issue?



Pairwise fit compatibility metric.